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Basic Indicators

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### Basic Indicators



#### 5.2 Market Internals

## Basic Indicators

In this unit, we will cover a variety of basic and advanced indicators that are unique to market internals. Given there are dozens, if not hundreds, of indicators based on these statistics, we will focus on those that candidates are most likely to run across and of which they should have a good working knowledge.

### Advance-Dcline Line

As noted in Level I, the advance-decline line (ADL) is a running total of the net number of stocks moving higher and lower. The line can be plotted with both daily and weekly data, and will typically show similar results. Again, be sure when analyzing weekly charts not to simply use daily data from Friday alone.

While some use the ADL to define bull and bear markets, it is also valuable as a confirming indicator used in conjunction with the popular averages, particularly with an intermediate-term time horizon. Points Y to Z in Figure 5.2.1 are a good example. It is also wise to combine the ADL with other indicators, either to provide support for the assessment or more pinpoint timing, such as with the stochastic, at point A on Figure 5.2.1.

The ADL can be extremely helpful in identifying major, cyclical turning points as well, though it often lags behind the popular indexes at bottoms. While this inconsistency can be frustrating, it accurately represents the market's tendency to top slowly and bottom quickly.

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**FIGURE 5.2.1** Daily bar chart of the Nasdaq 100 ETF, volume histogram, 13-day exponential average of net volume, advance-decline line, slow stochastic

## Up and Down Volume

As noted in Level I, up and down volume represent the number of shares traded on up and down ticks during the trading session. Like the ADL, it can also be shown as a “net” number (up volume minus down volume). Whether netted or individually, it is better when shown as an average, with Figure 5.2.1 showing a 13-day exponential average. Like the ADL, these measures can also be a good representation of the health of a market trend, with divergence analysis being key, as shown at points X, Y, and Z.

It should be noted that when looking for divergences between the indicator and in the indexes, it is important to monitor the group driving the trend. Specifically, when in an uptrend, it is the divergence in up volume (or net volume) that is important. This is because it is the bulls who are driving the trend, and it matters when they are running out of gas, not the bears. It is, of course, the opposite during downtrends, in that the technician should be watching for divergences in down volume for clues of a market bottom.

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Breadth Indicators

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## Breadth Indicators



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## Breadth Indicators

Another common way to analyze breadth data is by using indicators. A variety of them have been conceived over the years, some using raw data while others either use short-term moving averages on the results, or similarly, smooth the raw data before performing the calculations.

These indicators typically seek to gauge the short-term momentum of the market participants, with the assumption being that extreme readings occur at the start of trends, or when trends re-accelerate.

Notable examples include:

### Absolute Breadth Index

Developed by Norman Fosback in the '70s, this indicator is simply a 10-day simple moving average of the absolute value of advancers minus decliners. The theory being that one-sided markets are bullish, regardless of whether it is the bulls or bears who are in control. Advancing stocks will definitely outpace declining stocks at the start of a meaningful advance, while decliners will swamp advancers at washout bottoms.

Low readings, then, are the bearish ones, as they describe an indecisive market or so-called "split tape." This is a reliable market phenomenon to be aware of, and it is a concept also at the heart of many other indicators, such as Ari Wald's "sector culprit" indicator.

### Advance/Decline Ratio

This is the simple mathematical ratio of: (advancers / decliners), with notable levels being 5:1 and 10:1. Trading days showing heavy trading in one direction are viewed to be significant and the start of meaningful moves in the market.

### 80% and 90% Days

Similar to the simple ratios, breadth statistics can also be shown as a percentage of their total. Mathematically, this is expressed as: (advancers / (advancers + decliners)) and (decliners / (advancers + decliners)). An 80% day, for example, would be the same as a 4:1 ratio. Unchanged issues are typically ignored, and the same calculation can be performed on up and down volume. Trading days logging 80% or 90% of the activity in one direction can be significant and are a variation of "breadth thrusts," which will be explained later.

### Percentage of Stocks Above Their *n*-Day Average:

Yet another valuable way to describe the strength or health of a trend is to tabulate how many of the stocks are participating in

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above their 30-day average, then, is a good indicator of intermediate-term strength, while the percent trading above their 200-day average is a good indication of longer-term strength. Finally, these metrics are useful for sectors, broad market averages, and exchange-level data.

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Leadership Indicators

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## Leadership Indicators



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## Leadership Indicators

Like breadth, technicians have parsed, sorted, and manipulated data on new 52-week highs and lows (a.k.a. leadership) a wide variety of ways over the years. Also, like breadth indicators, these studies largely fall into several distinct categories, as follows:

## Raw Data

Unlike breadth, leadership statistics are extremely, and arguably most, useful merely in their “raw” form. Also, unlike breadth, highs and lows are not helpful when shown in cumulative form, such the ADL.

## Net Data

While new highs and new lows can be plotted on a chart separately, they are equally useful as a net number, with each print showing the value of (new highs - new lows). They can also be plotted as a percentage of the overall market as (new highs / total issues) and (new lows / total issues), or as a net ((new highs - new lows) / total issues).

Most importantly, leadership data is ideally suited for divergence analysis, and quite often marks meaningful tops and bottoms with pinpoint accuracy. This is true for both daily and weekly data.

## Percentage of New Highs

Finally, Gerald Appel, founder of the MACD, also invented another very valuable way to monitor leadership. His formula was to show the 10-day moving average of new highs divided by the total of highs plus lows (new highs / (new highs + new lows)). This ratio is intended to be used on exchange-level data and does a good job identifying both when a rally is weakening (when the indicator breaks below from above 70%) as well as good spots to buy bottoms (when the indicator moves back up through 30% from below).

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### Advanced Indicators



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## Advanced Indicators

### The Arms Index (TRIN Index)

Also called the Traders Index, the TRIN was invented by Dick Arms to measure how much volume is behind advancing and declining stocks. The formula is: (advancing stocks / up volume) / (declining stocks / down volume). The neutral value for the indicator is 1.

Readings less than 1 show heavier volume per advancing issue than declining volume per declining issue and are considered bullish. Readings greater than 1 show the opposite and are considered bearish. Technicians vary in how they use the data, with some plotting daily values, while others use a 10-day average of the data. It is also possible to calculate this as (advancing stocks / declining stocks) / (advancing volume / declining volume).

### McClellan Indicators

Both McClellan indicators were developed by Sherman and Marian McClellan.

The McClellan Oscillator measures the difference between a 19-day exponential moving average (EMA) of net advancers minus a 39-day EMA of net advancers. The results typically “oscillate” between +100 and -100, with it primarily being used as an overbought or oversold indicator. Like others of this kind, extreme readings (>100, or <-100) can indicate a powerful surge in momentum and potentially the start of a more sustainable trend.

The McClellan Summation Index is essentially an advance-decline line, or cumulative total, of McClellan Oscillator readings. Its goal is to be a better long-term indicator than the McClellan Oscillator, and more suited to identifying cyclical tops and bottoms.

### Zweig Breadth Thrust

Developed by Dr. Martin Zweig, the Zweig Breadth Thrust Indicator is a 10-day moving average of (advancers / (advancers + decliners)). Very similar to 80% and 90% days, this indicator tries to determine when momentum shift in the market is more sustained. Signals occur when the indicator moves from under 40% to over 62% and are fairly rare. They are commonly found at the start of meaningful (often cyclical) advances.

### Lowry's Reversals

Founded in 1938, Lowry's is the longest continuously running stock market research service in the U.S. A key part of their

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Award-winning paper.

The theory is that bear market bottoms are characterized by first a selling panic, and then a buying panic. Desmond uses 90% declining and down volume days to identify the selling panic. Once one is in place, the technician can start looking for either a 90% advancing and up volume day, or several 80% advancing and up volume days within a narrow span of time.

### 40:1 Ratio

Another indicator with the goal of marking panic bottoms is the 40:1 (pronounced “forty to one”) ratio or indicator, which measures the number of new lows to new highs (new lows / new highs). This indicator is binary in nature, giving either a “yes” or “no” reading. It often flashes “yes” in concert with 90% down days, as that is typically when new lows outpace new highs by 40 to 1 or more.

### High-Low Logic

Another indicator developed by Norman Fosback in the ‘70s, high-low logic builds on the absolute breadth index and is defined by taking the lesser of new highs and new lows on the day and dividing that number by total issues traded. Given the indicator uses the lesser of highs or lows, it usually has a very low reading, often less than 1%. Its goal is to identify when the market becomes “split” and posts a large number of both new 52-week highs and new 52-week lows (since it is the lesser figure that is used in the calculation).

### Hindenburg Omen

As its name implies, the Hindenburg Omen was designed to forewarn of market crashes. Proponents of the method will attest the indicator has indeed flashed a warning ahead of all meaningful market declines. Opponents will say that it has also flashed many warnings without a meaningful decline ever occurring.

The calculation is based on NYSE statistics and flashes a warning when all of the following occur:

1. Stocks are in an uptrend, as defined by the New York Stock Exchange Composite Index as having a rising 50-day average (many use the S&P 500 instead);
2. The McClellan Oscillator is negative, suggesting weak breadth; and
3. The high-low logic indicator is over 2.2%, which identifies a split and potentially vulnerable tape.

While we will leave it to candidates to research this method themselves before deciding whether to use it, it is undoubtedly a good example of just how far you can go in building indicators.



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Qualitative Analysis

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### Qualitative Analysis



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## Qualitative Analysis

While all the previously discussed indicators have specific formulas for their calculations, as well as defined buy and sell signals, there is also tremendous value in analyzing the market from a qualitative perspective.

### Range Shifts

One method is similar to how Connie Brown suggests using the Relative Strength Index (RSI). Specifically, she recommends monitoring the range in which the indicator trades, and that shifts in that range can be an early warning to trend changes in the market in question.

Technicians can apply that concept to indicators like the percentage of NYSE stocks above their 200-day average. Seeing the indicator slip below 40% while stocks are in a bull market suggests they are on an increasingly weak footing technically, and is often an early warning of an impending bear market.

### Fall Days and Spring Days

It can be valuable to describe market cycles as similar to the changing of the seasons.

“Spring” and “summer,” of course, refer to good times, or bull markets, while “winter” is the occasional bear market. It is also a convenient analogy, in that most of us enjoy four seasons, and winter lasts about one quarter of the time. Just like the average market cycle – three years up, and one year down, or thereabouts.

Just as the coming winter is typically foreshadowed by the first day you need to wear a sweater, the equity markets also often provide us “fall days” as a warning. These can be defined as the first meaningful surge of new 52-week lows in an established bull market – “established” meaning not the first year, at a minimum.

“Spring days,” on the other hand, are intended to identify emerging bull markets. They can be defined as when the percentage of stocks above their 50-day average moves back above 80% and when the percentage of stocks above their 200-day average moves back above 40%. In both cases, the indicators need to hold, and weekly exchange level data is useful for the calculations.

While these instances of market action are often present at cyclical turning points, they are qualitative in nature and subject to judgement and experience. With that said, it may be best to be on watch for a major top and to honor divergences after just one “fall day.” “Spring days” come in a pair, however,

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exact bottom, the entry will likely be close, and far more reliable.

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